Written Assignment - 2

Q1. Explain applications of linked list

A1. Linked lists act as extremely dynamic and efficient arrays if implemented correctly. They can be varible in size, allocating memory as and when required, allowing the array to be dynamically sized.
It can be of queues, stacks etc depending on its implementation. It can also ~~to~~ contain sorted, unsorted data ~~depening~~ depending on the user's need. Insertions in the middle as well as deletions ~~or~~ can be made very efficient.

Linked lists are used for ~~FIFO & LIFO~~ first in first out data structures and last in first out structures. They are also used in circular applications, undo and redo like used in editors for doubly linked lists.

Q2. Explain tree terminologies and describe ~~bina~~ binary tree representation using an array.

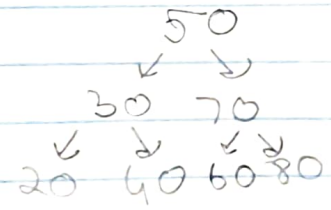
A2. A tree is a non-linear data structure with nodes connected by edges.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

A node is a basic element that contains data.
A root node is the first node of the tree with no parent node.
A parent node is a node which stores the child nodes and its own value
A ~~child~~ leaf node is a node with no child nodes ie. the array ends at them.
Child node - It is a node that is derived from another node.
Height of a tree - The max number of steps until the tree reaches a leaf node
Degree of Node - It is the max number of children of a node.

Binary tree means that each node can have upto two children.

Array - {50, 30, 70, 20, 40, 60, 80}



root node - 50
parent node e.g. 30 is parent of 20 and 40
child node e.g. 20, 40 are child nodes of 30
height of the tree is 3

Leaf nodes are 20, 40, 60 and 80 as they have no child nodes
This tree is configured to have the smaller numbers on the left and larger ones on the right

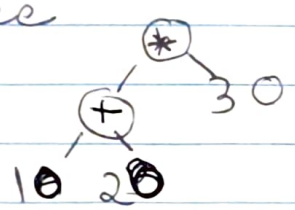
Q3. Short Note:
a. Representation of Expression Tree
b. Huffman coding

Ans. Representation of Expression Tree

Structure - Leaf nodes are numbers
Internal nodes are operators (+, -, x, ÷)

$(10+20) \times 30$

Tree



Each node contains data, pointer to left and pointer to right child.

It is stored as a binary tree and can be used for expression evaluation & conversion between infix, prefix and postfix.

b Huffman coding
It is an algorithm that takes input of an array of characters along with their frequency of occurances.
It is built from the bottom up, the nodes are sorted from minimum to maximum frequency
Extract 2 nodes of minumum frequency and create a parent node of them with a sum of their frequencies and add it to the sorted list of all the

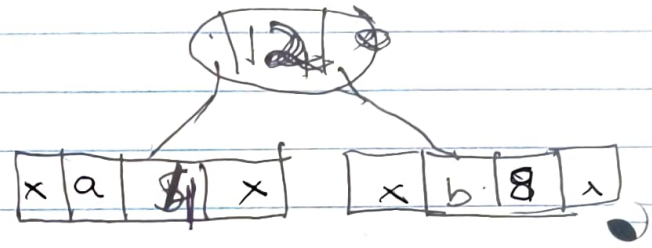
THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

nodes.

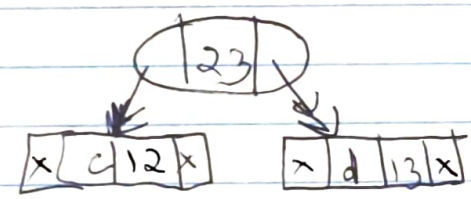
Repeat this step until the heap has only one node which becomes the root node. This creates a complete tree.

e.g.

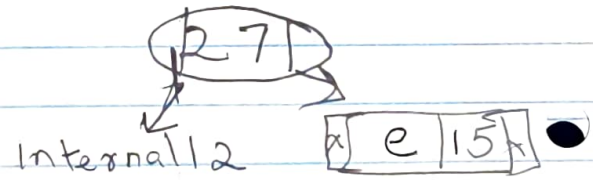
a 4
b 8
c 11
d 12
e 15
f 44



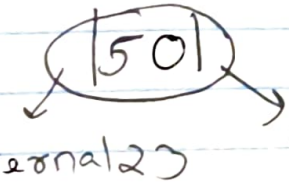
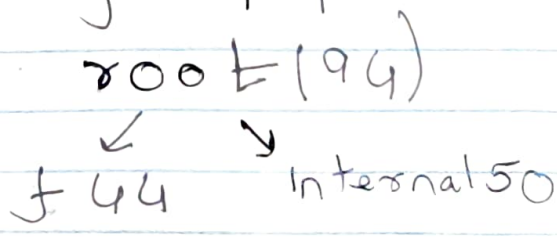
c 11
d 12
Internal 12
e 15
f 44



Internal 12
e 15
Internal 23
f 44



Internal 23
Internal 27
f 44



f 44
Internal 50